Computational Social Science with Images and Audio

Elliott Ash, Andrea Ciccarone, Philine Widmer

30 October 2026

Today, we turn to audio.

Pictures by Dall-E.



Figure: "The auditory sense of humans is important, symbolized"

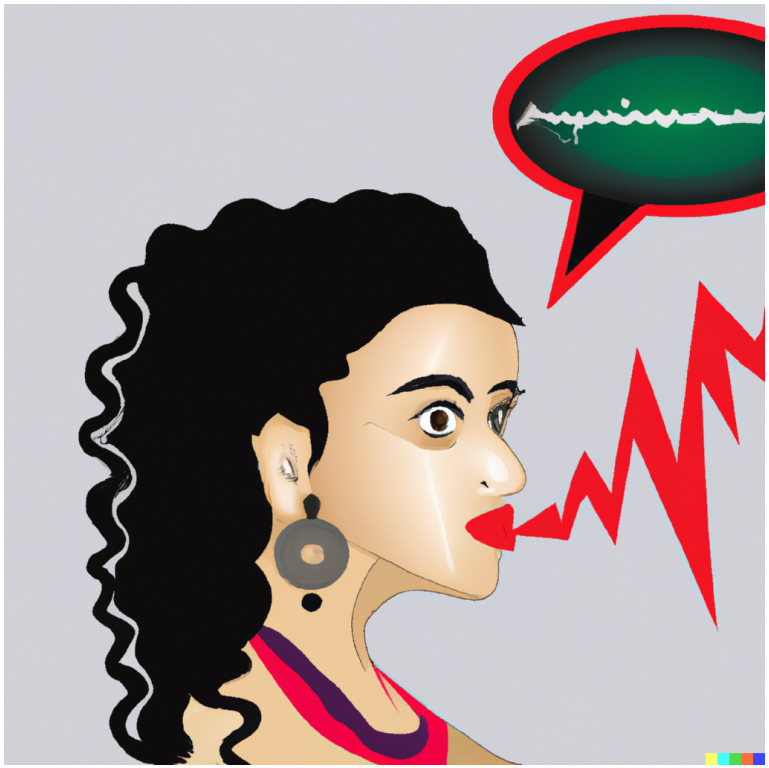


Figure: "Artwork on human speech"

Human hearing is interesting (not only for social scientists).

- Hearing is a critical sense for communication and survival
- As for vision, it enables us to interact with the world without physical contact
- Various areas of the brain are associated with auditory processing
- Humans are "analysts" and "generators" of audio at the same time
- Human hearing has its constraints
  - Limited frequency range: Humans typically hear between 20 Hz to 20 kHz, whereas some animals can perceive sounds beyond these limits
  - Difficulty in locating low-frequency sounds
- Human hearing is fascinating on several levels:
  - Fundamental: How does the auditory system decode sounds? $\rightarrow$ Beyond the scope of this class
  - Aggregate/social: What is the impact of sound on societal outcomes? $\rightarrow$ Central to our class

Audio analysis comes with many definitions.

- Many view it as a signal-processing challenge
- Julius O. Smith III: reconstructing auditory experiences
- In this class, we will explore techniques to analyze auditory data (recordings) in large quantities to study social science questions
  - Again, we begin where the auditory data already exists: we do not cover, from a technical perspective, the mechanisms of capturing this data (e.g., microphone technology)

What is sound?

- A sensory experience created by vibrations traveling through a medium (usually air)
- Like ocean waves move through water, sound waves propagate through the air
- As ocean waves interact with air, they create ripples in the atmosphere, analogous to **sound waves**

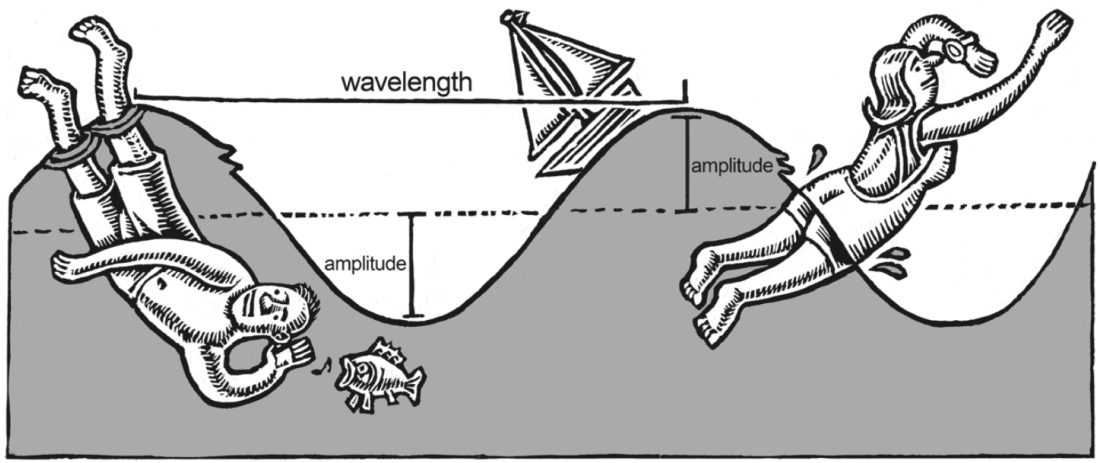


Figure: The analogy of ocean waves to sound waves; Source: Sulzer (2021)[1]

[1] Sulzer (2021). Music, math, and mind: The Physics and Neuroscience of Music. Columbia University Press.

Humans' hearing range is 20 Hz to 20 kHz.

- We begin to hear low vibrations at around **20 Hz**
- Teenagers can hear frequencies that extend up to about **20,000 Hz**, above which we humans do not perceive sound
- Each time you double the frequency, you move up by one octave
- Hence, with good hearing, we can hear across a ten-octave range:

1. 20 Hz
2. 40 Hz (20 Hz $\times$ 2)
3. 80 Hz (40 Hz $\times$ 2)
4. 160 Hz (80 Hz $\times$ 2)
5. 320 Hz (160 Hz $\times$ 2)
6. 640 Hz (320 Hz $\times$ 2)
7. 1280 Hz (640 Hz $\times$ 2)
8. 2560 Hz (1280 Hz $\times$ 2)
9. 5120 Hz (2560 Hz $\times$ 2)
10. 10240 Hz (5120 Hz $\times$ 2)
11. 20480 Hz (10240 Hz $\times$ 2)

Figure: Human hearing range

Hz, octave, and sampling frequency are important concepts around sound waves.

- Hz (Hertz): Measures the number of cycles per second of a periodic waveform
  - Higher Hertz means a higher-pitched sound
  - If a tone has a frequency of 440 Hz, the sound wave completes 440 cycles every second (A4)
- Octave: The interval between one musical pitch and another with half or double its frequency
  - 440 Hz and 880 Hz
- Sampling frequency: The number of samples per second from a continuous signal to make a digital signal
  - Standard CD sampling rate is 44.1 kHz...
  - ... meaning that the audio waveform is being sampled 44,100 times per second

The wavelength is an important sound wave characteristic.

- What is a sound wave, to begin with?
  - A transfer of sound energy through a medium
  - It is transmitted via the vibration of particles within that medium
- Illustration for non-physicists:
  - When something vibrates (like a speaker's diaphragm), it pushes on the neighboring air particles, increasing the pressure in that region $\rightarrow$ this push is then transferred from particle to particle $\rightarrow$ when it reaches someone's ear, the ear interprets the varying pressures as sound
- Wavelength: The physical distance between identical points in consecutive cycles of a sound wave
  - Determines the pitch

The amplitude is another important characteristic.

- Amplitude *in a general wave context*: the magnitude of change in the oscillating variable within the wave
  - $\rightarrow$ Larger amplitudes mean more energy and often translate to louder sounds when talking about sound waves
- Amplitude *in digital audio signal processing*: understood as instantaneous amplitudes that represent the "magnitude of change" in the pressure wave (the sound wave) from its equilibrium position at that particular moment
  - $\rightarrow$ The specific value of the audio waveform at a given sample point
  - What is the equilibrium position?

What does this sound like?

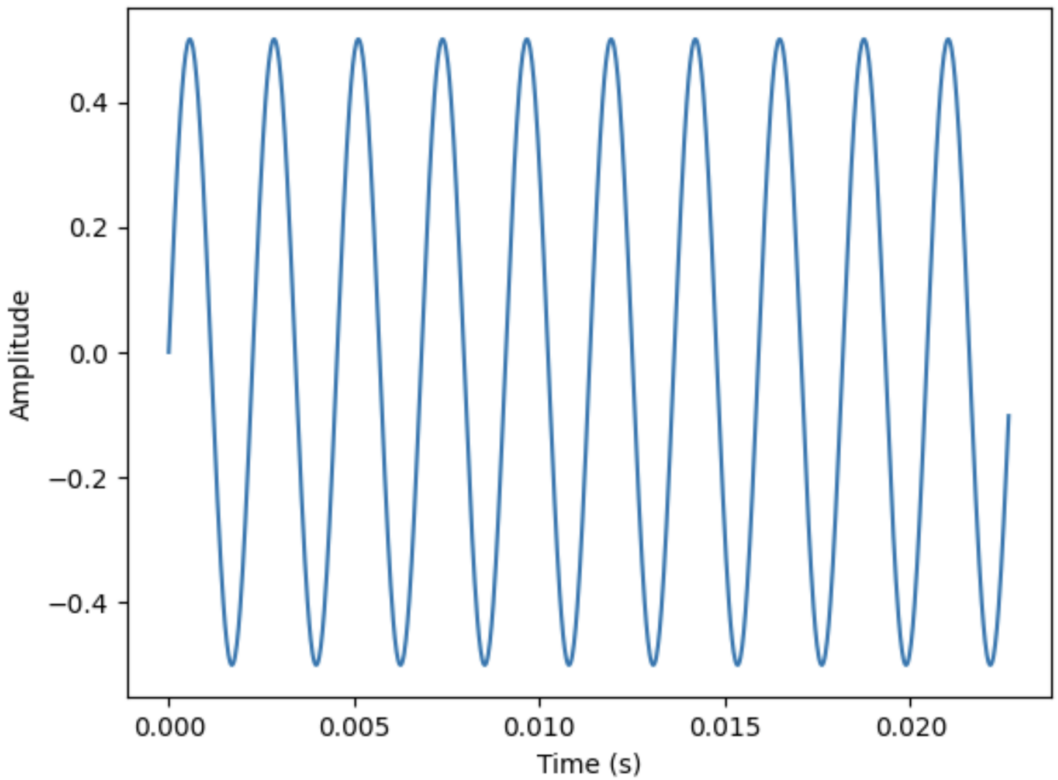


Figure: What sound could this be?

And what could this be?

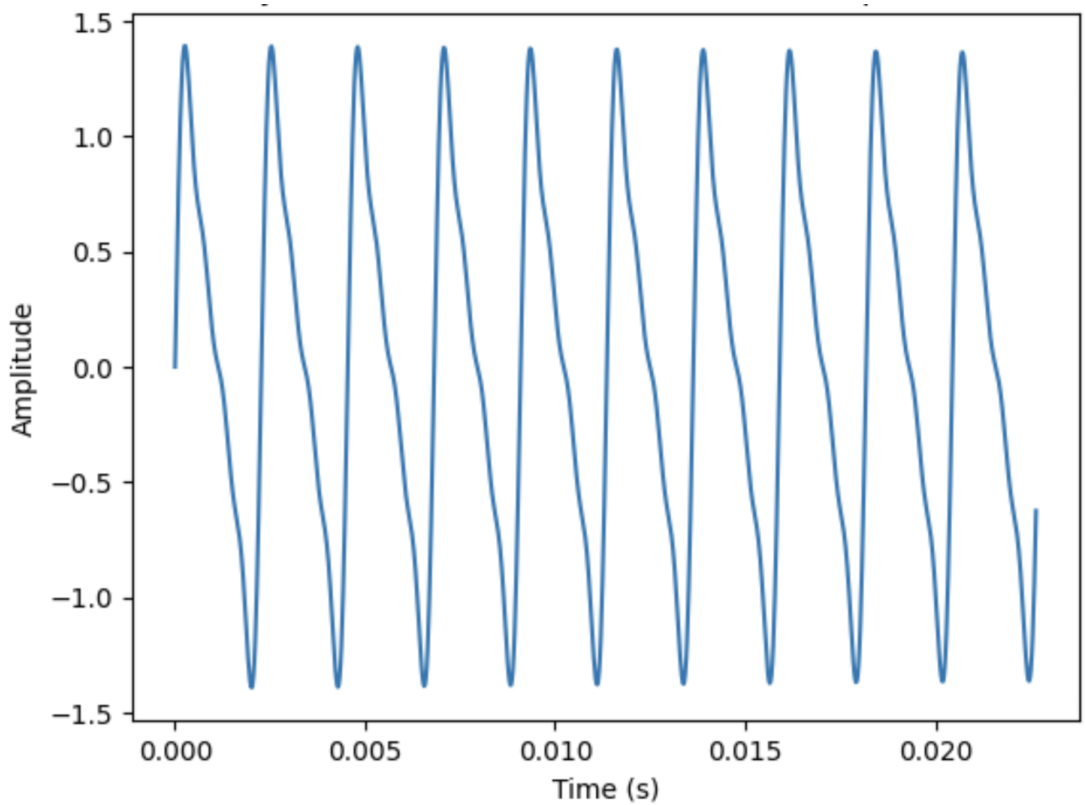


Figure: Any ideas?

This sound wave already looks more complex:

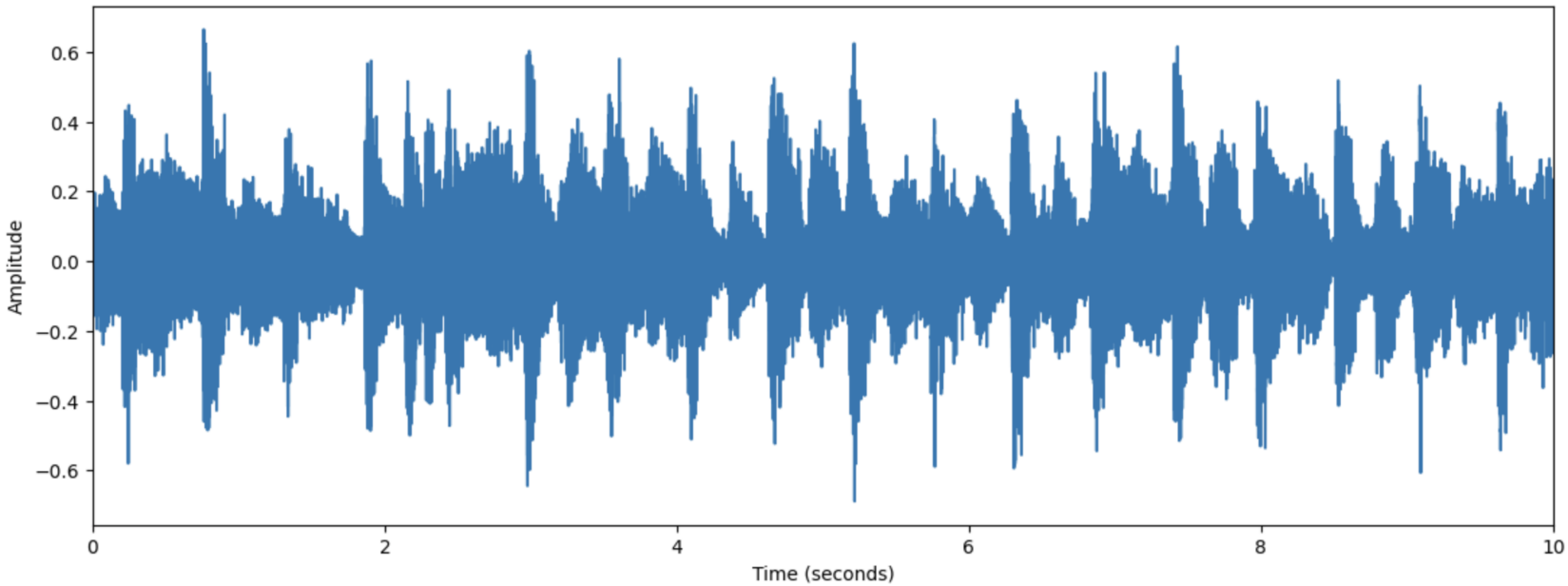


Figure: Hint: You certainly know this sound (too well).

Can we use the raw audio input for classification tasks?

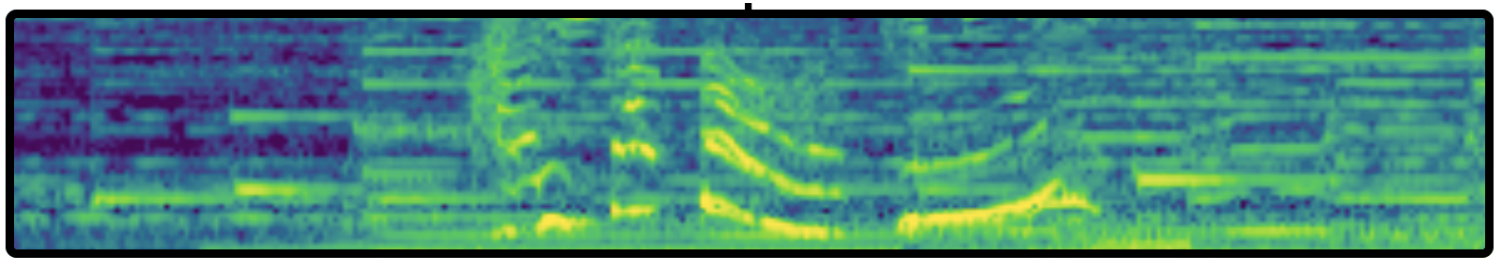
- Neural networks, such as CNN, can handle raw audio data $\rightarrow$ extract relevant features automatically (cf. images part)
- For very simple tasks, it may be possible with classical machine learning
  - Can you think of an example?
- More generally, audio waveforms are high-dimensional data
  - Computationally demanding: e.g., already 16,000 samples/second at 16kHz
  - Often not necessary $\rightarrow$ suitable format depends on task
- For both classical and neural approaches, feature extraction often useful

Modern approaches: Raw audio vs. learned features

- Trend in deep learning: Move toward raw audio or minimal preprocessing
- Spectrograms still dominant for efficiency
  - Spectrogram: Time-frequency representation (like frequency histogram at each time step)
  - Reduces sequence length: e.g., 100 frames/second (each 80 mel bins) vs. 16,000 samples/second
  - Sequence length matters for transformer
  - Used in modern speech recognition (e.g., Whisper)
- Hand-crafted features (MFCCs) are now often replaced by learned representations

Example: Log-mel spectrogram for speech recognition

- Modern speech recognition systems like Whisper use log-mel spectrograms
- Time-frequency representation fed as a sequence to the transformer



Log-mel spectrogram

Figure: OpenAI (2025)[2]

[2] https://openai.com/index/whisper/

Still one example for explicit feature extraction: Mel-Frequency Cepstral Coefficients (MFCCs).

- Humans perceive sound frequencies non-linearly (rather, logarithmically)
- Human sensitivity is greater to changes in lower frequencies
- The Mel scale is a way to measure pitch that matches how we hear sounds
  - It is a perceptual scale of pitches judged to be equal in distance from one another
  - Originally derived from experiments with human listeners
- MFCCs represent the power spectrum of an audio signal more in line with human hearing
  - They use the Mel scale

MFCCs capture the short-term power spectrum of sound.

- One typically begins by dividing the audio signal into short (overlapping) frames, for instance 20-40 ms
  - This allows assuming stationarity within each frame
  - Stationarity means that the statistical properties of the signal (like mean, variance) are constant over the frame's duration
- Several processing steps involved
- The bottom line is that we typically end up with 12-13 coefficients per frame
  - Empirically found to capture the most important features

With MFCCs, do we dimension-reduce?

- Assume an audio clip of 0.1 seconds (100 milliseconds)
- What's the dimension of the raw wave?
- How often can we take a 20ms frame with an overlap of, say, 50%?
- Now, each frame comes with 12-13 coefficients...

What models are used in audio analysis with deep learning?

The audio analysis landscape is diverse.

- CNNs are used for classification with audio data
  - Less prominently than in CV
  - Audio data is often transformed into spectrograms to use CNN
- Varied approaches depending on task
  - Speech recognition: Transformers on spectrograms (Whisper: see above)
  - Self-supervised learning: CNN + Transformer on raw audio (Wav2Vec 2.0, HuBERT)
  - Hybrid models combining CNN with RNN or Transformers
- Classical ML techniques still relevant for specific tasks
  - Recall explicit features (e.g., MFCC)
  - Similar considerations as for other unstructured data types (e.g., useful with limited labeled data)

Recommended reading: Mestre and Ryan (2026).

"Potential and Pitfalls of Audio as Data for Political Research"

- Strongly recommended reading for the exam[3]
- Data: all televised U.S. presidential debates from 1960 to 2020
- Features: pitch, energy, MFCCs, and embeddings (Wav2Vec 2.0)
- Applications:
  - Forced alignment: adding time stamps to existing transcripts
  - Speech characterization with pitch and energy
  - Speaker identification and emotion recognition with embeddings
- Pitfalls:
  - Recording conditions distort comparisons across speakers
  - Emotion recognition models can encode biases (e.g., gender)
  - Hence: compare within speakers, validate with human annotations

[3] Mestre, R., and Ryan, M. (2026). Potential and Pitfalls of Audio as Data for Political Research: Alignment, Features, and Classification Models. Political Analysis, 34(3), 312–328. Open access: https://doi.org/10.1017/pan.2025.10031. Replication code: https://github.com/rafamestre/audio-as-data